Name _____ Period _____ Date _____

4-2 Lesson Quiz: Graphing Rational Functions

short form

Question numbers match the Savvas Lesson Quiz. Questions 1 and 4 are not on this form.

2. Identify any vertical asymptotes of the graph of $y = \dfrac{(x+1)^2}{x^2 - 3x - 4}$.

Ⓐ $x = -4$ and $x = -1$

Ⓑ $x = -1$

Ⓒ $x = -1$ and $x = 4$

Ⓓ $x = 4$

Show your work.

The denominator factored:

_____

Where the denominator is 0, is the numerator 0 too?

_____

3. What are the vertical and horizontal asymptotes of the graph of $f(x) = \dfrac{3 - 2x}{4 - 3x}$?

Ⓐ $x = \frac{4}{3}$ and $y = \frac{2}{3}$

Ⓑ $x = \frac{2}{3}$ and $y = \frac{4}{3}$

Ⓒ $x = \frac{3}{4}$ and $y = \frac{2}{3}$

Ⓓ $x = \frac{3}{2}$ and $y = \frac{4}{3}$

Show your work.

Vertical asymptote (where is the denominator 0?):

_____

Horizontal asymptote (compare degrees, then leading coefficients):

_____

5. The graph of which equation below has a horizontal asymptote at $y = -3$?

Ⓐ $y = \dfrac{x - 3}{x + 1}$

Ⓑ $y = \dfrac{x}{2 - 3x}$

Ⓒ $y = \dfrac{1 - 3x}{2 + x}$

Ⓓ $y = \dfrac{1 + 3x}{2 - 3x}$

Write the horizontal asymptote of each choice.

A: $y =$ _____ B: $y =$ _____

C: $y =$ _____ D: $y =$ _____

Explain in one sentence how you found them.

_____

Scoring: Question 3 and Question 5 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

4-2 Lesson Quiz, short form: answer key

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2. D, $x = 4$**.** $x^2 - 3x - 4 = (x - 4)(x + 1)$, so the denominator is 0 at $x = 4$ and $x = -1$. At $x = 4$ the numerator is $(4 + 1)^2 = 25 \neq 0$: a vertical asymptote. At $x = -1$ the numerator is $(-1 + 1)^2 = 0$ too: $x + 1$ cancels ($y = \frac{x+1}{x-4}$ for $x \neq -1$), so the graph has a hole there, not an asymptote. Savvas quiz item 2, DOK 2, Realize skill F76. Taught by Example 2 (factor the denominator, then check the numerator). **Bonus 5:** 3 for D, 1 for the factored denominator, 1 for saying the numerator is also 0 at $x = -1$. Distractor C keeps both zeros of the denominator; B keeps only the zero that cancels; A factors with the wrong sign $(x + 4)$.

3. A, $x = \frac{4}{3}$ **and** $y = \frac{2}{3}$**.** $4 - 3x = 0$ gives $x = \frac{4}{3}$ (the numerator is $3 - 2 \cdot \frac{4}{3} = \frac{1}{3} \neq 0$). Same degree, so $y = \frac{-2}{-3} = \frac{2}{3}$. Savvas quiz item 3, DOK 1, Realize skill F76. Taught by Example 2 (set the denominator equal to 0, check the numerator) and Example 3 (Case 3, ratio of leading coefficients). **10 points:** 6 for A, 2 for the vertical-asymptote work, 2 for the horizontal-asymptote work. Distractor B swaps the two; C solves $4x - 3 = 0$; D uses the numerator.

5. C, $y = \dfrac{1 - 3x}{2 + x}$**.** Every choice has equal degrees, so the asymptote is the ratio of leading coefficients: A $\frac{1}{1} = 1$; B $\frac{1}{-3} = -\frac{1}{3}$; C $\frac{-3}{1} = -3$; D $\frac{3}{-3} = -1$. Savvas quiz item 5, DOK 2, Realize skill F76. Taught by Example 3 (Case 3). **10 points:** 6 for C, 3 for the four asymptotes (any three correct), 1 for the sentence naming degree and leading coefficients.

Items left off this form (available for a retake or Form B): Savvas item 1, which equation is shown in the graph (vertical asymptote $x = -2$, horizontal asymptote $y = 2$, right branch below $y = 2$; answer C, $p(x) = 2 - \frac{1}{2x+4}$: the denominator $2x + 4$ is 0 at $x = -2$, and at $x = 0$ C gives 1.75, below $y = 2$, while B gives 2.25; DOK 1, F20; taught by Example 4, asymptotes then a test point, and the $k + \frac{a}{x-h}$ form of Example 1, both cut); Savvas item 4, water added to 10 liters of a 12% salt mixture, $f(x) = \frac{1.2}{x+10}$, liters of water for a 1% mixture (answer 110 liters: $\frac{1.2}{x+10} = 0.01$ gives $x + 10 = 120$; DOK 2, F74; taught by Example 5, cut; the packet's Practice block is its parallel). The TE prints item 4's function as a "percent" but it gives a fraction (0.01 for 1%).

IXL replacement: SmartScore 70 or above on 7JJ, 79Q and UB5 replaces this quiz with 20/20.